

WEIS valuation methodology, v2: an empirically grounded bottom-up rebuild

WP-14 · tier internal · WEIS-2027 (this note is the evidence substrate; the tier-A WEIS artifact stays fiat-free and consumes only the dimensionless outputs of Sections 3-4) · 2026-07-14

The Privacy-is-Value Research Programme

2026-07-14

0. Verdict

The canonical point figures (678x, 31,000x) do not survive; the rebuild wins (role law; GR-3, GR-6). What survives is stronger than the point figures because it is now anchored in verified primary filings and primary experimental tables, and every quantity is an interval.

Three findings are load-bearing and new relative to draft-v1:

1. **Two independently-sourced brackets overlap in order of magnitude (this is NOT a triangulation).** The revealed annual flow the observing platform captures per user (Meta world-wide advertising ARPU, \$44.60 in FY2023, primary 10-K) and the median user's stated reservation value for the entire data stock, annualised over a chosen horizon (\$22.50-\$75/year across the plausible discount/tenure range, from the \$750 median WTA of Collis et al. 2022), both fall in the low-to-mid tens of dollars per year for the worldwide region and a 12-year-or-shorter horizon. The earlier "factor of about 1.6" headline is RETIRED. The two quantities are a seller's reservation value (a one-time stock) and a third party's gross revenue flow; no equilibrium theory predicts they coincide, and the subject-side number becomes a flow only after an annualisation knob whose setting moves it across the full \$22.50-\$833/year band (E30-E33, perpetuity $r=3\%$ through the declined-income subgroup). Their order-of-magnitude proximity for one region/horizon slice is a bracketing convenience, not corroboration; it is reported with the full annualisation-induced band beside it (GR-8), never as two families "agreeing".
2. **An endowment effect is present; its magnitude anchor is CORRECTED.** In the one experiment in the set that puts WTA and WTP on the same choice (Acquisti, John and Loewenstein 2013), the elicited monetary constants are \$10 anonymised versus \$12 identified (E34) and \$10 versus \$14 (E35): a WTA:WTP ratio of VALUATIONS of about 1.2 to 1.4. A separate statistic from the same experiment, 52.1% choosing privacy when endowed with it versus 9.7% when they must buy it (E36), is a ratio of PARTICIPATION PROPORTIONS (5.37), not a ratio of monetary valuations, and v1 mis-imported it as the endowment ratio anchor, inflating the reported effect roughly fourfold. The endowment RATIO is re-anchored on the monetary wedge (~ 1.2 -1.4) actually elicited; the 5.37 participation ratio is retained only as separate corroboration that an endowment effect exists in direction, never as its magnitude.

3. **The aggregation wedge is re-derived on a data-attributable, stock/stock basis, and its super-additive claim collapses under returns-to-scale evidence (Run-3 tightening).** Two quantities were conflated in v2's single

- w. (a) The single-record MODELLING/TARGETING premium, taken like-for-like within the broker market (general demographic record E19 \$0.0005 to targeted segment E20 \$0.0021, $\sim 4x$), is order $\sim 10^0$. This is a cross-sectional price-LEVEL ratio between two record TYPES for one subject's record; it does NOT measure super- or sub-additivity of AGGREGATING many subjects' data, and NO figure in the set directly measures aggregation super-additivity. Independent returns-to-scale evidence (Bajari et al. 2018, E46-E47: forecast error falls only at the diminishing $1/\sqrt{N}+1/\sqrt{T}$ rate, with genuine but diminishing gains in history-length T and data variety N robustly flat-to-negative, i.e. super-additivity in variety is affirmatively REJECTED; Varian E49: a 0.1% subsample suffices for the estimation task, so the marginal analytic value of additional volume is ~ 0 at scale; third-party broker profiling accuracy is low, E50) shows any aggregation term SATURATES rather than compounds, and points to order $\sim 10^0$. (b) The observer-to-broker realised-value GAP, now built STOCK/STOCK (lifetime data-attributable capture \$107-\$214 over a one-time targeted broker record \$0.0021, worldwide central band), is order 10^4 to 10^5 and no longer crosses the flow/stock boundary L132 flagged; but the same returns-to-scale evidence shows this gap is NOT aggregation value, it is atomisation discount plus market-position rent plus lifetime accumulation. v2's characterisation of w as "super-additive, order 10^{1-105} " is CORRECTED: the word "super-additive" is DROPPED entirely (no figure demonstrates it and E47 rejects it); what survives is a single-record modelling premium of order $\sim 10^0$ that saturates; the large gap is rent, not aggregation, and it must not harden into a point (GR-8).

The market mechanism through which the value is realised or lost is named in Section 4 and is concrete: the real-time-bidding programmatic auction and the data-broker resale market are where behavioural data actually clears, and the subject is a counterparty in neither.

Run-3 tightening note (2026-07-15)

Run-3 does not re-open what survived Run-2's three hostile lenses (L132). It tightens exactly two objects the hostile verifier left wide.

1. **Observer capture is now DATA-ATTRIBUTABLE, not gross.** v2 carried gross ARPU as the proxy for data-attributable surplus and flagged the fraction as unmeasured. Run-3 brackets the targeting-premium fraction x from external literature (E39-E45): x in $[0.04, 0.52]$, central band $[0.2, 0.4]$, policy-clean midpoint ~ 0.20 (ATT aggregate). The proxy carried downstream is $x \times$ ARPU, not ARPU. Worldwide, that is \$8.92-\$17.84/yr central versus \$44.60/yr gross (Section 4.4). NARROWED: the data proxy drops to $0.2x$ - $0.4x$ of the v2 figure; gross is kept only as an upper bound. NOT narrowed: which single x is correct stays open, and the $\sim 13x$ gap between the independent $\sim 4\%$ (E39) and Google's own $\sim 52\%$ (E44) is unresolved (GR-8).
2. **The wedge is split and its super-additive claim collapses.** v2's single w (order 10^{1-105}) built its top by dividing the observer's GROSS ANNUAL flow by a ONE-TIME broker record price, crossing a flow/stock boundary (L132). Run-3 (i) fixes the boundary by building the top STOCK/STOCK (lifetime data-attributable capture \$107-\$214 over a one-time targeted record

\$0.0021, worldwide central) and (ii) applies returns-to-scale evidence (E46-E50: Bajari et al.'s $1/\sqrt{N}+1/\sqrt{T}$ convergence, genuine-but-diminishing gains in T, and flat-to-negative variety effect N; Varian's finding that a 0.1% subsample suffices, evidencing ~ 0 marginal analytic value at scale; low broker profiling accuracy) to show any aggregation term saturates and points to order $\sim 10^0$. **NARROWED:** the SUPER-ADDITIVE claim collapses entirely, from $[10^{1,10}5]$ to a single-record modelling premium of order $\sim 10^0$ (the word "super-additive" is dropped; no figure demonstrates it and E47 rejects it); the top is now dimensionally clean; the data-attributable fraction shaves the central top from $\sim 4.5 \times 10^5$ to $\sim 1 \times 10^5$. **NOT narrowed** (and now correctly labelled): the raw observer-to-broker gap remains order $10^{4-10}5$, non-identified, but is rent and atomisation discount and accumulation, NOT aggregation value (Section 4.3).

3. **Untouched:** the s-share intervals (Section 4.1), the WTA:WTP endowment ratio (Section 4.2), and the no-triangulation headline (Section 5) are left as Run-2 set them. The data-attributable fraction bears on s only through $1/x$, which does not change its order and is noted in place (Section 4.4).

1. Why a bottom-up rebuild, and what the figures are NOT

Every dollar figure in Section 2 is **gross advertising (or ad-resale) revenue or a stated-preference elicitation**. No primary source in the set decomposes platform revenue into the portion attributable specifically to behavioural-data targeting versus reach, creative, and brand. Therefore:

- ARPU is the flow the observer captures per user. GROSS ARPU is an **upper bound** for the data-attributable surplus, not the "value of a person's data", and is no longer used as the data proxy. Run-3 partially closes the data-attributable measurement gap: the observer-capture proxy carried downstream is **x x ARPU**, where x is the targeting-premium fraction, x in $[0.04, 0.52]$, central band $[0.2, 0.4]$, policy-clean midpoint ~ 0.20 (Section 4.4, E39-E45). What remains open is which single x within that band is correct; the fraction is bounded, not point-identified (GR-8).
- No Google/Alphabet per-user figure is asserted. Alphabet's 10-K discloses neither ARPU nor a user/MAU count (grep-verified zero hits in the primary filing); any circulating Google "revenue per user" is an analyst division and is marked UNVERIFIED, not carried.
- Meta retired ARPU/MAU reporting after Q1-2024 (switched to ARPP on Daily Active People). FY2023 is the last clean full-year ARPU vintage; the figures are not comparable to post-2024 ARPP.

The point-figure failure modes from draft-v1 (Section 1 there) still hold and are not repeated: ill-posed near-zero denominator, aggregation dependence, and the non-arithmetic reframing of the high-end figure. This note's job is the positive construction the referee asked for.

2. Evidence table (all fiat figures live here, GR-3)

Every row is a CITED EXTERNAL MEASUREMENT with source, year, units, and flow/stock class. None is a model output; none asserts a value of privacy.

2.1 Revealed observer capture (advertising flow per user)

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E1	44.60	USD / Face-book+Messenger MAU / year, worldwide	FY2023	per-year flow	Meta 2023 Form 10-K, ARPU section (SEC EDGAR CIK 1326801, acc. 000132680124000038)	VERIFIED-primary
E2	13.12	USD / MAU / quarter, worldwide	Q4-2023	per-quarter flow	same 10-K, ARPU table	VERIFIED-primary
E3	68.44	USD / MAU / quarter, US & Canada	Q4-2023	per-quarter flow	same 10-K, ARPU-by-region	VERIFIED-primary
E4	23.14	USD / MAU / quarter, Europe (incl. RU, TR)	Q4-2023	per-quarter flow	same 10-K	VERIFIED-primary
E5	5.52	USD / MAU / quarter, Asia-Pacific	Q4-2023	per-quarter flow	same 10-K (position + 11x consistency check)	VERIFIED-primary
E6	4.50	USD / MAU / quarter, Rest of World	Q4-2023	per-quarter flow	same 10-K	VERIFIED-primary
E7	226.93	USD / MAU / year, US & Canada	FY2023	per-year flow	sum of quarterly regional rows (Meta method); annual regional total not printed in 10-K	VERIFIED-secondary
E8	75.57	USD / MAU / year, Europe	FY2023	per-year flow	as E7	VERIFIED-secondary

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E9	20.04	USD / MAU / year, Asia-Pacific	FY2023	per-year flow	as E7	VERIFIED-secondary
E10	15.83	USD / MAU / year, Rest of World	FY2023	per-year flow	as E7	VERIFIED-secondary
E11	131,948	USD millions, total advertising revenue	FY2023	per-year flow	Meta 2023 10-K income statement (~98% of total revenue)	VERIFIED-primary
E12	134,902	USD millions, total revenue	FY2023	per-year flow	Meta 2023 10-K	VERIFIED-primary
E13	237,855	USD millions, Google advertising total	FY2023	per-year flow	Alphabet 2023 10-K (SEC EDGAR CIK 1652044, acc. 000165204424000022, goog-20231231.htm)	VERIFIED-primary
E14	175,033	USD millions, Google Search & other	FY2023	per-year flow	same Alphabet 10-K	VERIFIED-primary
E15	31,510	USD millions, YouTube ads	FY2023	per-year flow	same Alphabet 10-K	VERIFIED-primary
E16	31,312	USD millions, Google Network	FY2023	per-year flow	same Alphabet 10-K	VERIFIED-primary
E17	307,394	USD millions, Alphabet total revenue	FY2023	per-year flow	same Alphabet 10-K (ad ~77% of total)	VERIFIED-primary

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E18	(none)	Google per-user / ARPU	—	—	Alphabet 10-K discloses NO ARPU and NO MAU (grep = 0 hits)	UNVERIFIED - dropped

2.2 Data-broker resale and RTB spot prices (per-record / per-impression)

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E19	0.0005	USD / person, general demographic (= \$0.50 per 1,000)	2013	per-record, one-time	FT “How much is your personal data worth?” (Steel et al.); reprinted fastcompany.com/2682323	VERIFIED-secondary
E20	~0.0021	USD / person, targeted segment (e.g. auto-buyer; = \$2.11 per 1,000)	2013	per-record, one-time	same FT interactive	VERIFIED-secondary

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E21	~0.005	USD / data attribute	2013	per-record, one-time	FT interactive (pay-walled); fastcompany reprint 403'd; NOT independently confirmed against Steel et al. primary	UNVERIFIED - dropped from numeric use
E22	80.28	Chinese fen, training-set total winning-price CPM (x1000 = CPM)	2013	per-impression clearing (RTB), training partition	Zhang, Yuan, Wang, Shen, arXiv:1407.7073 Table 3 (iPinYou training partition)	VERIFIED-primary (native unit, training-set winning price); USD conversion UNVERIFIED
E23	1 to 5	USD CPM, open exchange	c.2023-24	per-impression price	Choozle CPM benchmark (choozle.com/blog/umar, cheat-sheet)	indicative (single-vendor benchmark, unconfirmed)
E24	5 to 15	USD CPM, private marketplace	c.2023-24	per-impression price	same	indicative (single-vendor benchmark, unconfirmed)
E25	10 to 25+	USD CPM, programmatic guaranteed	c.2023-24	per-impression price	same	indicative (single-vendor benchmark, unconfirmed)

Note on E22: native figure only, and specifically the TRAINING-partition total winning-price CPM (winning/paying price x1000 in fen), not a market-wide average; the earlier “avg market/paying

clearing price” gloss conflated market price with paying price and omitted the train/test split. No dollar equivalent is asserted. Order of magnitude cross-checked only indicatively by E23-E25 (single-vendor benchmark, unconfirmed): open-exchange \$1-5 CPM = \$0.001-\$0.005 per impression. This cross-check is order-of-magnitude only and is not a model input.

2.3 Stated-preference valuations (WTA / WTP)

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E26	750	USD, median WTA (entire Facebook data stock), YouGov N=4,149	2021	one-time stock	Collis, Moehring, Sen, Acquisti, WEIS 2022 (weis2022.econinfosec.org ...weis22-collis.pdf, p.9)	VERIFIED-primary
E27	1,000	USD, median WTA, DDP sample	2021	one-time stock	same, Section 5	VERIFIED-primary
E28	73	USD / year (authors’ annualisation of E26 over 12-yr tenure)	2021	derived per-year flow	same (authors’ own derivation)	VERIFIED-primary (derived)
E29	6.1	USD / month (authors’ derivation)	2021	derived per-month flow	same	VERIFIED-primary (derived)
E30	1,000 (White) / 500 (Black)	USD, median WTA by race	2021	one-time stock	same, heterogeneity section	VERIFIED-primary
E31	1,000 (Men) / 558 (Women)	USD, median WTA by sex	2021	one-time stock	same (CORRECTS v1’s \$600 female figure)	VERIFIED-primary

#	Figure	Units	Year	Flow/stock	Source (re-solvable)	Confidence
E32	600	USD, median WTA, lowest-income group (DDP)	2021	one-time stock	same (the true source of the mis-attributed \$600)	VERIFIED-primary
E33	10,000	USD, median WTA, income-declined group	2021	one-time stock	same (highest subgroup)	VERIFIED-primary
E34	\$10 anon vs \$12 identified; \$2 wedge	USD, experimental design constants (Exp 2 real cards)	2013	per-decision	Acquisti, John, Loewenstein, JLS 42(2):249-274 (heinz.cmu.edu/~acquisti/papers/...JLS-2013.pdf)	VERIFIED-primary
E35	\$10 vs \$14; \$4 wedge (Exp 1 arm)	USD, design constants	2008/2013	per-decision	same	VERIFIED-primary
E36	52.1% (WTA) vs 9.7% (WTP); ratio 5.37	share of subjects choosing privacy	2013	per-choice proportion	same (Exp 2)	VERIFIED-primary

2.4 Named comparators not asserted numerically (GR-8/GR-9)

#	Comparator	Flow/stock	Status
E37	Prince & Wallsten, "How much is privacy worth around the world?"	per-month WTA by data type	named comparator; no verified figure in this note's input; A4 to source before any numeric use
E38	Savage & Waldman, "Privacy tradeoffs in smartphone applications"	per-app WTP	named comparator; A4 to source before numeric use

These two are recorded so the flow-class landscape is complete and the note does not silently imply the four measured families are the only ones. Their specific figures are UNVERIFIED here and are not carried.

2.5 Data-attributable fraction (targeting premium) and returns to scale (Run-3)

Two evidence families added in Run-3. The first (E39-E45) measures the fraction of gross advertising value attributable to behavioural data, so observer capture can be written $x \times$ gross rather than gross. The second (E46-E50) bounds the super-additivity of aggregation. All figures are dimensionless ratios; the one dollar figure (E43) is a CITED EXTERNAL MEASUREMENT, never an asserted value of privacy (GR-3). Publisher-side and advertiser-side premia are DIFFERENT quantities and are never averaged together (GR-8).

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E39	~4% (reported ~\$0.00008/impression)	percent uplift in publisher revenue per cooked vs uncooked impression	2019	AGGREGATE (publisher- realized average)	Marotta, Abhishek & Acquisti, “Online Tracking and Publishers’ Revenues”, WEIS 2019 (millions of RTB trans- actions, one large US media company) [https://weis2019.econinfosec.org/wp-content/uploads/sites/6/2019/05/WEIS_2019.pdf]	VERIFIED- secondary
E40	52% revenue gap for opt-out im- pressions; 0.23% opt-out incidence	percent revenue gap (targetable vs opted-out); percent opt-out incidence	2020	MARGINAL (selected opt-out minority)	Johnson, Shriver & Du, “Consumer Privacy Choice in Online Ad- vertising”, Marketing Science 39(1):33- 51 [https://pubsonline.informs.org/doi/10.1287/mksc.39.1.33]	VERIFIED- secondary

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E41	51% marginal price premium (trackable vs untrackable); ~20% aggregate Apple-user ad-revenue decline under ATT	percent price premium; percent aggregate revenue decline	2023-24	MARGINAL (51%) + AG-GREGATE (~20%)	Kraft, Bleier, Skiera & Koschella, "Granular Control and Privacy Decisions: Evidence from Apple's ATT", SSRN 4598472 [https://papers.ssrn.com/sol3/papers.cfm	VERIFIED-secondary

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E42	-18% (desktop EU 2016, 42M impr / 111 pubs) and -23% (mobile in-app 2023, 218M impr / 10,526 apps) ad- impression price without tracking	percent price decrease	2023	AGGREGATE (controlled price regression)	Laub, Miller & Skiera, “The Economic Value of User Tracking for Publishers”, arXiv 2303.10906 / SSRN 4251233 [https://arxiv.org/abs/2303.10906]	ATTRIBUTION VERIFIED; MAGNI- TUDE PENDING (au- thor/venue/direction consistent, but the -18%/-23% split and the impres- sion/publisher/app leg is NOT yet independ- ently confirmed against the primary table; A4/A12 to confirm verbatim before this controlled- regression leg is treated as settled, GR-9)

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E43	+31% cost per incremental customer without offsite tracking data (median \$38.16 -> \$49.93); implies ~1-1/1.31 = ~0.24 data-attributable (superseding published version)	percent CPA increase; USD per incremental customer (cited external)	2025	AGGREGATE (advertiser-side)	Wernerfelt, Tuchman, Shapiro & Moakler, "Estimating the Value of Offsite TRACKING Data to Advertisers: Evidence from Meta", NBER WP 32765 / SSRN 4915952, Marketing Science 2025 (~107,000 advertisers) [https://www.nber.org/papers/w32765]	VERIFIED-secondary (published version supersedes the 2022 working paper SSRN 4198438, which reported +37%/\$43.88->\$60.19/~0.27; the newer +31%/~0.24 is used, and the central x-band [0.2,0.4] is unaffected either way)
E44	Top-500 publishers avg -52% / median -64% revenue without third-party cookies	percent publisher revenue decrease	2019	MARGINAL, DOWN-WEIGHTED (vendor conflict)	Ravichandran & Korula (Google), "Effect of disabling third-party cookies on publisher revenue" (RCT, Google Ad Manager) [https://services.google.com/fh/files/misc/party_cookies_publisher_revenue.pdf]	VERIFIED-secondary (first-party vendor study; ~13x above E39, reported with equal prominence GR-8)

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E45	Contextual eCPM ~36% lower than behavioural	percent CPM difference	2019-20	(dropped)	Dentsu Aegis / GumGum industry study [https://gumgum.com/industry-study-proves-the-effectiveness-of-contextual-over-behavioral-targeting]	UNVERIFIED - dropped (vendor-authored, direct conflict; not load-bearing)
E46	Relative forecast error converges to its floor at rate $1/\sqrt{N}+1/\sqrt{T}$; marginal value of each data point decreases to zero as data size grows	dimensionless convergence rate	2018	returns-to-scale (SUB-additive in quantity)	Bajari, Chernozhukov, Hortacsu & Suzuki, "The Impact of Big Data on Firm Performance", NBER WP 24334 (Assertion 2 / Implication 1) [https://www.nber.org/papers/w24334]	VERIFIED-primary

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E47	History-length T effect on big-error probability ~ -0.35 in the no-trend model, flattening to ~ -0.06 under time fixed effects (genuine but diminishing forecast gains in T); data-variety N effect “robustly flat”, insignificant, negative in ≥ 2 specs	change in probability (dimensionless)	2018	returns-to-scale (T delivers real diminishing gains; super-additivity in VARIETY N REJECTED)	Bajari et al., NBER WP 24334 (AR1-AR2 magnitudes as printed; all-groups GR2)	VERIFIED-primary
E48	Naive-Bayes classifier keeps improving with diminishing returns even past $\sim 500M$ entries (the strongest pro-scale case, still diminishing)	data entries (count)	2013	returns-to-scale (diminishing at extreme volume)	de Fortuny, Martens & Provost, “Predictive Modeling with Big Data: Is Bigger Really Better?”, Big Data 1(4):215-226 (cited in Bajari et al.)	VERIFIED-secondary

#	Figure	Units	Year	Class	Source (re-solvable)	Confidence
E49	Google reportedly powers decision-support using only 0.1% subsamples	fraction of data used	2014	revealed marginal value ~0 at scale	Varian, “Big Data: New Tricks for Econometrics”, JEP 28(2):3-28 (cited in Bajari et al.)	VERIFIED-secondary
E50	Two-attribute broker profiling ~24.4% correct (range 12.9-32.3%); gender alone ~random (~50%); single-attribute improvement over random 0-77%	percent correctly identified / percent improvement over random	2019	low aggregation accuracy (bounds modelling premium)	Neumann, Tucker & Whitfield, “How Effective Is Third-Party Consumer Profiling?”, Marketing Science 38(6):918-926 [doi 10.1287/mksc.2019.1188]	VERIFIED-secondary

Reading the fraction (E39-E45). The evidence splits into two regimes that must not be conflated. (A) The AVERAGE realized uplift behavioural data adds to a publisher’s actual revenue is small: ~4% (E39). (B) The MARGINAL impression-level premium (a trackable impression vs an otherwise-identical untracked one) is large and consistent at ~50-52% (E40, E41, E44), with controlled price regressions and aggregate ATT revenue loss landing at ~18-27% (E41, E42, E43). The marginal ~50% cluster OVERSTATES the aggregate data-attributable fraction because it is a conditional/selection comparison struck in an equilibrium where most other impressions remain tracked (E40’s 0.23% opt-out incidence is itself the selection warning, and is a negative-side fact: revealed demand to opt out is tiny). The AGGREGATE family (~0.04 publisher-realized; ~0.18-0.23 controlled regression E42, magnitude PENDING independent confirmation; ~0.20 ATT revenue loss; ~0.24 implied advertiser-side, E43 superseding published version) is the correct family for scaling observer capture. **Defensible central band x in [0.2, 0.4]** (anchored by E41 ATT ~0.20 and E43 advertiser ~0.24; E42’s controlled-regression leg is held as corroboration pending confirmation, and the band does not depend on it), bounded below by ~0.04 (E39) and above by ~0.52 (E40/E44 marginal, hostile-verifier stress), with ~0.20 (E41 ATT aggregate) as the policy-clean midpoint. NEGATIVES (GR-8): the ~13x divergence between the independent E39 (~4%) and Google’s own E44 (~52%)

is unresolved and undercuts any single point estimate; the largest marginal figures come from parties with a commercial stake (Google E44, Meta-affiliated data E43, GumGum E45); publisher-side and advertiser-side premia are different quantities; all estimates are context- and era-specific (2016-2024, RTB display vs in-app vs social).

Reading returns to scale (E46-E50). The empirical record says aggregation value SATURATES rather than compounds. Forecast error falls only at the diminishing $1/\sqrt{N}+1/\sqrt{T}$ rate; history-length T delivers genuine but diminishing forecast gains, while the VARIETY dimension N is robustly flat-to-negative, i.e. super-additivity in variety is affirmatively REJECTED (E46-E47); even the most favourable “bigger is better” evidence shows only diminishing marginal gains past 500M entries (E48); a firm using only a 0.1% subsample for a given estimation task reveals ~ 0 marginal analytic value of additional volume at scale (Varian, E49; note this is a subsample-suffices claim, not a data-disposal claim: the firm retains the full corpus and uses it elsewhere); and broker profiling on combined attributes is barely better than chance (E50). No figure in the set directly measures aggregation super-additivity, and E47 rejects it in the variety dimension; there is no empirical term that would carry an aggregation wedge to order 10^2+ . This bounds the single-record modelling premium (w_{super}) to order $\sim 10^0$ (Section 4.3).

3. Reconciling the incommensurable measures (no silent averaging)

The figures above measure **five different quantities**. They are not interchangeable and are never averaged across categories. Stated explicitly:

Measure	What it prices	Native unit / class	Cannot be compared to X without...
Observer capture (E1-E10)	gross ad revenue the platform realises per user	per-year flow	...scaling by the data-attributable fraction x in $[0.2, 0.4]$ central (E39-E45, Section 4.4); gross alone overstates the data proxy
Broker resale (E19-E21)	spot price to buy one atomised record	per-record, one-time stock	...an aggregation model to reach modelled value
RTB clearing (E22-E25)	price of one ad impression at auction	per-impression	...impressions/user/year to reach a per-user flow
Stated WTA (E26-E33)	compensation demanded to surrender the whole stock	one-time stock	...an annualisation assumption (tenure or discount rate)
Endowment wedge (E34-E36)	anonymised-vs-identified pricing gap / WTA-WTP proportion gap	per-decision / ratio	...nothing; it is already dimensionless as a ratio

Conversion assumptions, stated once and used consistently:

- **A-STOCK-TO-FLOW.** To compare a one-time WTA (E26) with an annual observer flow (E1), annualise the stock. Two defensible routes: (i) authors' amortisation over the observed 12-year tenure, giving $E28 = \$73/\text{year}$ (note simple division $\$750/12 = \62.50 ; the authors' $\$73$ embeds a present-value method, so the two differ by the discount assumption); (ii) perpetuity annuitisation at rate r : $\text{flow} = \$750 \times r$. See Section 5 for the r -sweep.
- **A-REGIME.** The observer flow (E1) is realised gross advertising turnover under the *extraction* regime; the WTA (E26) is a *reservation value* (the compensation the subject would demand to surrender the stock) in a *bargaining* regime. These are NOT two prices of one good from opposite sides of one market: one is a seller's outside-option floor, the other a third party's gross revenue flow. No equilibrium theory predicts a reservation value equals a gross-revenue flow; they would coincide only under competitive pass-through with the subject as counterparty, a condition this note explicitly says does not hold (Section 6, "the subject is a counterparty in neither"). Their numerical proximity is therefore a bracketing convenience, NOT corroboration, and the earlier claim that the overlap is "informative rather than tautological" is WITHDRAWN. A triangulation claim would have to supply the missing pass-through/bargaining model that makes reservation $\sim$ gross flow and show it plausible; absent that, this note reports two independently-sourced brackets that happen to overlap in order of magnitude, nothing more.
- **A-NO-CROSS-AVERAGE.** The five families are never pooled into one mean. The headline band (Section 5) is a *bracketing* of two like-classed quantities (annual observer flow; annualised subject reservation), not an average over all five.

4. The dimensionless model quantities, with intervals (discharges M4)

Every quantity below is an interval; each endpoint traces to a figure. This is the referee-requested content that draft-v1 dropped.

4.1 Surplus-appropriation share s

Normalise the appropriable per-subject data surplus over one period to 1. The subject captures s ; the observer captures $1 - s$. Dimensionless by construction (GR-3 satisfied).

Regime	Interval	Endpoint provenance
s (extraction)	$[\sim 10^{-5}, 0.05]$, dimensionless	Lower/anchor: the share must be (subject's annual realisation) / (annual appropriable surplus), BOTH per-year. So the numerator uses the broker resale price times the annual resale multiplicity f (records clearing per year): $s \sim f \times$ (broker record price E19-E20, $\sim \\$0.0005$-$\\0.0021) / (observer annual flow E1, $\\$44.60/\text{yr}$). v1 set the numerator to a one-time record price over an annual flow, which carries units of yr^{-1} and is NOT a share; that endpoint is CORRECTED here by making the numerator an annual flow via f. With f of order 1 to 10 (unmeasured), $s \sim 10^{-5}$ to 10^{-4}; the anchor moves with f, which is the residual non-identification and is stated, not hidden. Upper 0.05 = an UNMEASURED generous allowance for uncaptured consumer surplus from the free service (the zero-price side of the two-sided market; not data-attributable; stated as a gap, GR-8), NOT a measured figure.

Regime	Interval	Endpoint provenance
s (sovereign, projection)	[0.3, 0.7]	Nash bargaining split with roughly symmetric outside options once a credible outside option exists (IEEE 7012, E4-C10). 0.5 = symmetric Nash; 0.3 = firm-favourable asymmetry; 0.7 = subject-favourable. This is a bargaining-theoretic PROJECTION, not a measurement; no deployed sovereignty market has been observed to price s (GR-8). Soft cross-check only: annualised median subject reservation (E28 \$73/yr, or the wider \$22.50-\$75/yr band) is of the same order as observer capture E1 (\$44.60/yr), consistent with a subject able to command a non-trivial share once at the table. Per Section 3 A-REGIME a reservation value and a gross-revenue flow are not commensurable, so this corroborates direction, not magnitude.

The “appropriation gap” is the movement of s from the extraction interval to the sovereign interval: a shift from order $10^{-4/-5}$ up to order 10^{-1} , i.e. three to four orders of magnitude in the subject’s captured share, with BOTH endpoints now dimensionless shares of the same per-period surplus, expressed as a share change and never as a ratio to a near-zero base. Because both s values are shares of one normalised per-period surplus, GR-3 dimensionlessness holds at the numeric endpoints, not merely “by construction” of the normalisation; the corrected extraction anchor (annual flow over annual flow, with the resale multiplicity f explicit) is what makes this true. This is the honest residue of “678x”. The number 678 is not reconstructed.

4.2 WTA:WTP endowment ratio

Quantity	Interval	Anchor	Provenance
WTA:WTP endowment ratio (value-based)	[~1.2, ~4]	~1.2-1.4	Primary VALUE anchor: the elicited monetary constants \$10 vs \$12 (E34) and \$10 vs \$14 (E35) give a WTA:WTP ratio of valuations of ~1.2-1.4 (AJL 2013). The upper reach to ~4 brackets the larger endowment effects reported across the broader privacy endowment-effect literature (NOT in this verified set; flagged for A4 to source before any tighter claim). The 5.37 figure from E36 is a ratio of PARTICIPATION PROPORTIONS (52.1% vs 9.7% choosing privacy), NOT a ratio of monetary valuations; it is cited below as direction-only corroboration and is explicitly NOT the anchor.

The corrected value-based ratio (~1.2-1.4, reaching higher in the broader literature) sits at or above the ~1-2 typical of ordinary private goods, and the participation-rate gap (5.37, E36) independently signals that far more subjects will hold privacy than will buy it. Both are demand-side signatures of the appropriation problem: the subject values holding privacy above the price at which she would buy it. But the magnitude of the value wedge is modest in this verified set (~1.2-1.4); the strong effect (5.37) is in participation, not in elicited dollars, and the two are not interchangeable. v1 conflated them.

4.3 Aggregation wedge w (re-derived, Run-3)

v2 reported a single w spanning order 10^1 to 10^5 whose top divided the observer's GROSS ANNUAL flow by a ONE-TIME broker record price, crossing a flow/stock boundary (L132). Run-3 separates the two quantities v2 conflated and fixes the boundary crossing.

Quantity	Interval	Provenance / basis
w_super = single-record modelling/targeting premium, like-for-like within the broker market (same time basis, stock/stock); a price-LEVEL ratio, NOT an aggregation-returns measurement	order $\sim 10^0$, saturating (NOT super-additive)	Anchor: intra-market segment-price LEVEL premium, general demographic record E19 (\$0.0005) to targeted segment E20 (\$0.0021) = $\sim 4\times$ (order $10^{0.6}$). This is a cross-sectional ratio between two record TYPES for one subject's record; it does NOT measure super- or sub-additivity of aggregating many subjects' data, and NO figure in the set directly measures aggregation super-additivity. Bounded by returns-to-scale evidence, NOT extrapolated: forecast error falls only at $1/\sqrt{N}+1/\sqrt{T}$, T gains are genuine but diminishing and variety N is flat-to-negative with super-additivity affirmatively rejected (E46-E47), extreme-volume gains are diminishing (E48), a 0.1% subsample suffices for a given estimation task so marginal analytic value is ~ 0 at scale (E49), combined-attribute broker profiling is barely above chance (E50), and the marginal data-attributable premium is $\sim 0.2-0.5$ (E41-E43). No empirical term compounds this beyond order $\sim 10^0$.

Quantity	Interval	Provenance / basis
appropriation/rent gap = lifetime data-attributable observer capture / one-time atomised broker record (STOCK/STOCK)	order 10^4 to 10^5, non-identified	Numerator: lifetime data-attributable capture (a STOCK) = $x \times \text{ARPU} \times \text{tenure}$ = [0.2,0.4] $\times$ \$44.60/yr $\times$ 12yr = \$107-\$214 (worldwide, central band; Section 4.4). Denominator: one-time targeted broker record E20 (\$0.0021, like-for-like with a modelled profile), a STOCK. w = \$107-\$214 / \$0.0021 = 5.1×10^4 to 1.0×10^5 . Both terms are stocks; the flow/stock boundary L132 flagged is removed.

How the flow/stock crossing is fixed. v2's top was (annual flow) / (one-time stock), dimensionally (yr^{-1}) and inflating. Run-3's top is (lifetime accumulated stock) / (one-time stock): the observer's data-attributable take accumulated over a subject's tenure, divided by the one-time price the atomised record fetches. Both are stocks, so the ratio is a clean dimensionless number. Applying the data-attributable fraction x also lowers the central top from v2's $\sim 4.5 \times 10^5$ (gross, US-and-Canada) to $\sim 1 \times 10^5$ (data-attributable, worldwide), about two-thirds of an order.

Is the upper reach super-additive, or does it saturate? It saturates, and no figure even measures super-additivity. The returns-to-scale evidence (E46-E50) is decisive: the observer-to-broker gap of order 10^{4-105} is NOT the product of super-additive aggregation. It decomposes into (i) w_{super} , the single-record modelling/targeting premium, order $\sim 10^0$ (and even this is a price-LEVEL ratio, not an aggregation-returns measurement); (ii) the ATOMISATION DISCOUNT and MARKET-POSITION RENT (the broker resale market prices an atomised record 4-5 orders below its realised lifetime value precisely because neither the subject nor small buyers are the clearing venue, Section 6); and (iii) lifetime accumulation (the $\sim 12x$ tenure factor). Only (i) is a data-value premium; (ii) and (iii) are rent and accumulation. **v2's "w is super-additive, order 10^{1-105} " is CORRECTED:** the word "super-additive" is dropped entirely (no figure in the set demonstrates it and E47 affirmatively rejects it in the variety dimension); the single-record modelling premium is order $\sim 10^0$; the large gap is rent, not compounding aggregation.

What survives, stated plainly. The economically meaningful wedge a subject-side aggregator could re-capture as VALUE-ADD (w_{super}) is narrow, order $\sim 10^0$, and bounded by hard returns-to-scale evidence; no figure in the set even measures aggregation super-additivity, so the label is dropped. The full observer-to-broker gap (order 10^{4-105}) is real and now dimensionally clean (stock/stock), but it is the appropriation gap (rent extracted because the subject is not admitted to the clearing venue), not evidence of super-additive data value. Both remain non-identified as point values (GR-8): w_{super} because its anchor is a price-level ratio and its ceiling is a bound not a measurement, the rent gap because it moves with region, tenure, and which broker price anchors the denominator (stress: US-and-Canada, $x=0.4$, general record \$0.0005 pushes it to $\sim 2 \times 10^6$; worldwide, $x=0.04$, targeted record holds it near 1×10^4). The interval is NARROWER than v2's in the sense that matters: the

super-additive claim has collapsed entirely (from $[10^{1,10}5]$ to a single-record modelling premium of order $\sim 10^0$), and the residual bigness is now correctly attributed to rent.

4.4 Data-attributable observer capture (Run-3)

v2 carried GROSS ARPU as the observer-capture proxy and flagged the data-attributable fraction as an open gap. Run-3 replaces gross with **x x gross**, where x is the targeting-premium fraction from E39-E45.

Quantity	Interval	Provenance
x = data-attributable (behavioural-targeting) fraction of gross ad value	[0.04, 0.52], central band [0.2, 0.4], policy-clean midpoint ~ 0.20	Lower ~ 0.04 : publisher-realized average uplift (E39, Marotta et al.). Central band: ATT aggregate revenue loss ~ 0.20 (E41, Kraft et al.), advertiser-side ~ 0.24 implied from +31% CPA (E43, Wernerfelt et al., superseding published version; the retired 2022 working paper gave +37%/ ~ 0.27), with controlled price regressions ~ 0.18 - 0.23 (E42, Laub et al.) as corroboration whose magnitude is PENDING independent confirmation. Upper ~ 0.52 : marginal auction premium / opt-out gap (E40, E44), a selection-conditional MARGINAL figure that overstates the aggregate and is held as a hostile-verifier stress.
data-attributable annual observer capture (worldwide) = x x E1	central \$8.92-\$17.84/yr; full range \$1.78-\$23.19/yr	x-band x \$44.60/yr (E1). Midpoint $\sim \$8.92/\text{yr}$ at $x=0.20$. Dollar figures are labelled conversions of the dimensionless x against a cited external ARPU (GR-3); the MODEL output is x itself.

Relative to the gross figure used in v2, scaling by x shaves the data-attributable proxy to roughly $0.04x$ - $0.52x$ of gross (central $0.2x$ - $0.4x$): gross is retained only as an upper bound. This is the proxy carried into the wedge numerator (Section 4.3) and the mechanism discussion (Section 6). It does NOT change the ORDER of the extraction share s (Section 4.1): using data-attributable surplus rather than gross as the denominator of s raises s by $1/x$ (a factor of ~ 2.5 - 5), which leaves s at order $10^{4/-5}$; that endpoint is left as in v2, with this factor noted.

5. Per-person annual value: per-category ranges, NOT a single headline (conversions in-table)

The user asked for a defensible real-number range. The honest answer is that triangulation does NOT support a single headline range, and per-category ranges are given instead. This is the more honest outcome, and it is the one the adversarial commensurability pass forces (Section 3, A-REGIME). GR-3 requires the model's asserted OUTPUT to be dimensionless; the only dimensionless outputs that survive are the share-shift and the wedge-order of Section 4, NOT a "convergence" number.

What is NOT asserted (RETIRED): the v1 claim that two measurement families "converge to within a factor of about 2" is withdrawn. A seller's reservation value (a one-time stock) and a third party's gross-revenue flow have no theoretical reason to coincide (Section 3, A-REGIME); the subject side becomes a flow only after an annualisation knob whose setting moves the median anchor across the full \$22.50-\$833/year band below. Landing near the observer flow is a property of a chosen region/horizon slice, engineered by that free discount/tenure choice, not an independent measurement agreeing. The overlap is reported as a coincidental order-of-magnitude proximity for one slice, with the full annualisation-induced band beside it (GR-8).

Asserted (dimensionless) outputs: (i) the surplus-appropriation share s moves from order 10^{-4} to order 10^{-1} (extraction) to order 10^{-1} (sovereign projection), a three-to-four order share-shift (Section 4.1); (ii) the aggregation wedge splits (Run-3): the single-record modelling premium w_{super} is order $\sim 10^0$ and SATURATES (returns-to-scale, E46-E50; "super-additive" is dropped, no figure demonstrates it and E47 rejects it), while the observer-to-broker realised-value gap is order 10^4 to 10^5 , stock/stock, non-identified, and is rent plus atomisation discount plus accumulation, NOT aggregation value (Section 4.3); (iii) the data-attributable fraction of observer capture is x in $[0.2, 0.4]$ central, $[0.04, 0.52]$ full (Section 4.4); (iv) the WTA:WTP value wedge is ~ 1.2 - 1.4 with a participation-rate endowment signal of ~ 5 (Section 4.2). No single per-person dollar-per-year figure is asserted as "the value of privacy".

Per-category ranges (cited external measurements + labelled conversions; NOT triangulated, NEVER pooled into one mean):

Category	Low	Anchor	High	Conversion assumption (in-table)
Observer GROSS annual capture (upper bound only)	\$15.83/yr (E10, Rest of World)	\$44.60/yr (E1, worldwide)	\$226.93/yr (E7, US & Canada)	none beyond regional selection; gross ad flow, retired as the data proxy, kept as an upper bound

Category	Low	Anchor	High	Conversion assumption (in-table)
Observer DATA-ATTRIBUTABLE annual capture (Run-3)	\$1.78/yr (x=0.04, worldwide)	\$8.92/yr (x=0.20, worldwide)	\$23.19/yr (x=0.52, worldwide)	x x E1; x in [0.04,0.52] targeting-premium fraction (E39-E45); central band \$8.92-\$17.84/yr (x in [0.2,0.4])
Annualised subject reservation (median WTA)	\$22.50/yr (E26 \$750 x r=3%)	\$62.50-\$73/yr (E26 \$750; simple /12yr = \$62.50; authors' PV = E28 \$73)	\$75/yr (E26 \$750 x r=10%)	one-time WTA annualised; the horizon/discount knob ALONE spans this row (perpetuity r=3% to r=10%), so the anchor is a chosen slice, not a datum
Annualised subject reservation (full subgroup band)	\$42/yr (E30 Black \$500 / 12yr)	-	\$833/yr (E33 declined-income \$10,000 / 12yr)	one-time subgroup WTA / 12-yr tenure; reported with equal prominence (GR-8) so the reader sees the band straddles the observer flow on both sides
Broker resale (one-time record)	\$0.0005 (E19)	-	\$0.0021 (E20)	one-time per-record price; E21 dropped (UNVERIFIED)
Endowment wedge (value-based)	~1.2 (E34)	-	~1.4 (E35)	ratio of elicited monetary valuations; the 5.37 of E36 is a participation ratio, cited separately, not here

No single headline range (dimensionless): per-category, both the observer annual capture and

the median-WTA annualised reservation happen to fall in order 10^1 to 10^2 USD/year for the worldwide region and a 12-year-or-shorter horizon, but this overlap is coincidental (Section 3, A-REGIME) and slice-dependent: it holds for r in $[5\%, 10\%]$ and worldwide, and fails for $r \leq 3\%$ (WTA falls below the observer flow), for the US & Canada region (observer flow rises to \$227/yr), and for the declined-income subgroup (WTA rises to \$833/yr). Neither category is “the value of privacy”, and the model does NOT assert they agree. The surviving dimensionless outputs are the s -shift and the w -order of Section 4, not a headline dollar-per-year number.

5.1 Sensitivity of the band

The band moves under three assumptions. Each row shows the median anchor shifting; endpoints scale similarly.

Assumption swept	Setting	Annualised subject reservation (from E26 \$750)	Observer capture used
Discount rate r (A-STOCK-TO-FLOW route ii, perpetuity \$750 $\times r$)	$r = 3\%$	\$22.50/yr	—
	$r = 5\%$	\$37.50/yr	—
	$r = 10\%$	\$75.00/yr	—
	authors' PV method (E28)	\$73/yr	—
	simple /12yr	\$62.50/yr	—
Which observer measure	worldwide (E1)	—	\$44.60/yr
	Europe (E8)	—	\$75.57/yr
	US & Canada (E7)	—	\$226.93/yr
Does wedge w accrue subject-side?	no (extraction, $s \sim 10^{-4}$)	subject realises ~broker price, ~\$0.001/yr-equiv	observer keeps ~\$44.60/yr
	yes (sovereign, s in $[0.3, 0.7]$)	subject realises 0.3-0.7 of surplus; at $s=0.5$ and observer-flow normalisation, order \$22-\$113/yr	observer keeps remainder

Reading the sensitivity: the sweep is the EVIDENCE that the overlap is a chosen slice, not a discovered fact. The annualised median subject reservation moves from \$22.50/yr ($r=3\%$) through \$62.50-\$75/yr (simple /12yr and $r=10\%$) while the observer measure moves from \$44.60/yr (worldwide) to \$226.93/yr (US & Canada): a stock can be annualised to match almost any target flow by choosing the horizon, so the fact that one slice (worldwide observer, r in $[5\%, 10\%]$) overlaps is a property of that slice, engineered by the free discount/tenure choice. At $r \leq 3\%$ the subject reservation falls below the worldwide observer flow; at the US & Canada region the observer flow sits above the entire median-WTA band. The single largest mover is **which observer measure** (a factor of ~ 14 from Rest

of World to US & Canada), and the second is the **discount/horizon knob** (a factor of ~ 3.3 from $r=3\%$ to $r=10\%$); between them they straddle the observer number on both sides, which is exactly why no single headline point is asserted. Whether the wedge accrues subject-side (extraction vs sovereign regime) moves the subject's realised share by the three-to-four order share-shift of Section 4.1.

6. The market mechanism (named, load-bearing)

The role's failure mode is a monetary claim with no market mechanism. The mechanism is concrete and has a pricing venue that actually exists:

Where behavioural data clears today. The real-time-bidding programmatic advertising auction (E22-E25) and the data-broker resale market (E19-E21) are the two spot markets in which behavioural data is priced. The subject is a counterparty in **neither**. The observer aggregates, models, and sells access, and captures the annual flow (E1-E10); the subject's only realisable market price is the broker resale price, 4-5 orders of magnitude below the flow. This is why s is near zero in the extraction regime (Section 4.1): not because the subject values privacy at zero (she does not, E26), but because she is not admitted to the market where her data clears.

How the share s moves. The consent interface is the bargaining protocol that sets s . Notice-and-consent is a take-it-or-leave-it offer whose outside option is service exclusion, which drives the subject's equilibrium share to zero (a mechanism result, not a moral one). IEEE Std 7012-2025 propose-and-respond (subject proposes terms; organisation accepts, negotiates once, or declines; bilateral signed record) inverts the offer direction and supplies a credible outside option, moving s into the bargaining interval (E4-C10).

Where the wedge w is captured. A subject-side aggregator / data coalition (the data-as-labour market instrument, Arrieta-Ibarra et al. 2018; Posner and Weyl 2018) aggregates individual behavioural streams on the subject side and sells access, moving the wedge w (Section 4.3), or a negotiated fraction of it, back to subjects instead of the observer.

Why the market does not unravel. Personal-data markets fail on adverse selection toward a lemons equilibrium (Akerlof 1970; Spiekermann et al. 2015). A non-transferable, non-purchasable, Sybil-resistant quality stake (E4-C06, register C42, carried with its proof obligation) is the quality signal that lets the market of the two paragraphs above price quality without unravelling. This is a design claim and an open conjecture, not a measured result.

What is lost, stated plainly. In the extraction regime the subject's share s is near zero and the wedge w accrues entirely to the observer. The sovereignty the thesis describes has a market price, that price is currently paid to someone else, and the venue where it is paid is the RTB auction and the broker resale market named above.

7. $M(u,y)$ market-maturity gate (unchanged from v1; A3 owns entry)

The proposed operationalisation carries forward from draft-v1 Section 5 unchanged: $M : [0,1] \times R_+ \rightarrow [0,1]$, $M(u,y) = u(1 - e^{-y/\tau})$, with u = fraction of counterparties able to price sovereignty and y = market depth. As a product factor, realised value = $M(u,y) \times$ potential value, so $u = 0$ (no counterparty can price sovereignty) yields zero capture, which is exactly the extraction-regime observation (Section 6): the subject is not admitted to the pricing venue, so $M \sim 0$ today. A3 owns whether this form enters WP-14.

8. Limits (GR-8)

- **Data-attributable fraction is bounded, not point-identified (Run-3).** The fraction x attributable specifically to behavioural targeting is now bracketed by external literature at x in $[0.04, 0.52]$, central band $[0.2, 0.4]$, policy midpoint ~ 0.20 (E39-E45). What remains open is which single x is correct: the $\sim 13x$ divergence between the independent publisher-realized $\sim 4\%$ (E39) and Google's own $\sim 52\%$ (E44) is unresolved, and publisher-side vs advertiser-side premia are different quantities. Gross ARPU is retired as the data proxy and kept only as an upper bound; the proxy carried downstream is $x \times \text{ARPU}$. Two provenance corrections in this finalisation: E43 now cites the superseding published version (NBER w32765 / Marketing Science 2025, $+31\% \rightarrow \sim 0.24$; the 2022 working paper SSRN 4198438 $+37\%/\sim 0.27$ is retired), and E42's $-18\%/-23\%$ split is **ATtribution-VERified** but **MAGNITUDE-PENDING** independent confirmation against the primary table (A4/A12), so its controlled-regression leg is corroboration only and the central band $[0.2, 0.4]$ does not depend on it.
- **Sovereign-regime s is a projection.** $[0.3, 0.7]$ is bargaining theory, not measurement. A deployed sovereignty market that fails to move s despite a working 7012 protocol and a functioning stake is reported with equal prominence as a confirmation (GR-8).
- **The wedge is re-derived and its super-additive claim is withdrawn (Run-3).** The single-record modelling/targeting premium w_{super} is order $\sim 10^0$ and SATURATES (bounded by returns-to-scale evidence E46-E50, not extrapolated); its anchor is a cross-sectional price-LEVEL ratio (E19- $\rightarrow$ E20 $\sim 4x$), NOT an aggregation-returns measurement, and NO figure in the set directly measures aggregation super-additivity, which E47 affirmatively rejects in the variety dimension. The word "super-additive" is dropped entirely; v2's "super-additive, order 10^{1-105} " is withdrawn. The full observer-to-broker realised-value gap (order 10^{4-105}) is now a clean STOCK/STOCK comparison (lifetime data-attributable capture over one-time record), so the flow/stock crossing L132 flagged is removed, but that gap is rent plus atomisation discount plus accumulation, not aggregation value. Both remain non-identified as points; the rent gap moves with region, tenure, and denominator choice (stress range $\sim 1 \times 10^4$ to $\sim 2 \times 10^6$).
- **Vintage mismatch.** Broker prices (E19-E21) and the RTB dataset (E22) are 2013 and historical; observer ARPU (E1-E10) is FY2023; stated WTA (E26-E33) is 2021. Cross-vintage comparison may understate current broker prices and is a source of the per-category bands' width, not a point estimate.
- **Meta metric discontinuity.** FY2023 is the last clean ARPU vintage; not comparable to post-2024 ARPP.
- **Google per-user dropped, not estimated (E18):** no primary basis exists.
- **iPinYou USD conversion dropped (E22):** native fen figure only, and specifically the training-partition winning-price CPM.
- **E21 (\$0.005/attribute) dropped as UNVERIFIED:** the FT interactive is paywalled and the reprint 403'd; it is not used as the wedge lower-bound endpoint, which is re-derived from E19- $\rightarrow$ E20 ($\sim 4x$, rounding to order 10^1).
- **The extraction share is dimensionless only with an explicit resale frequency.** $s(\text{extraction})$ is $(\text{subject annual realisation})/(\text{annual surplus})$; the numerator is the broker record price times the unmeasured annual resale multiplicity f . v1's numeric endpoint divided a one-time stock price by an annual flow (units yr^{-1} , not a share); the corrected anchor states f and moves with it. The order (10^{-5} to 10^{-4}) is robust for f of order 1 to 10.
- **The endowment ratio anchor was corrected.** v1 anchored the WTA:WTP ratio on a participation-proportion ratio (5.37, E36), which is not a ratio of valuations and inflated the

effect ~fourfold; the value-based anchor is ~1.2-1.4 (E34/E35). The participation ratio is retained only as direction-only corroboration.

- **No triangulated headline.** The v1 “two families converge to within a factor of ~2” claim is retired. A reservation-value stock and a gross-revenue flow have no theoretical reason to coincide; their overlap for one region/horizon slice is coincidental and is reported with the full annualisation-induced band (\$22.50-\$833/yr) beside it (GR-8). Per-category ranges replace the single headline.
- The M(u,y) form, the Sybil-resistance signal (C42), and reputation appreciation (E4-C07) are open conjectures / proposals, not measured returns.

9. Handoff

- **A5 (economist-referee):** the M4 finding is discharged in Section 4 (every quantity is now an interval with figure-traced endpoints) and Section 5.1 (explicit sensitivity sweep). Rate the rebuild.
- **A2 (prose port):** the tier-A WEIS artifact consumes ONLY the dimensionless outputs of Sections 4-6 (share intervals, endowment ratio, wedge order, per-category ranges, named mechanism); the v1 convergence/triangulation framing is retired and must not be ported. Apply the S5 patch (weis_seventh_capital_S5_patch.md) for the 5.1/5.2 replacement. It must not carry the Section 2 dollar table; that stays in this internal note.
- **A4 (bib):** add Collis et al. (WEIS 2022); confirm AJL (2013), ATW (2016), Arrieta-Ibarra et al. (2018), Posner-Weyl (2018), Spiekermann et al. (2015), Akerlof (1970), IEEE 7012-2025; source Prince-Wallsten and Savage-Waldman (E37-E38) before any numeric use. **Run-3 additions (E39-E50):** Marotta, Abhishek & Acquisti (WEIS 2019); Johnson, Shriver & Du (Marketing Science 2020); Kraft, Bleier, Skiera & Koschella (SSRN 4598472, ATT); Laub, Miller & Skiera (arXiv 2303.10906, cite as attribution-verified/magnitude-pending until the primary -18%/-23% table is confirmed); Wernerfelt, Tuchman, Shapiro & Moakler (cite the superseding published version NBER WP 32765 / SSRN 4915952 / Marketing Science 2025, +31%; the 2022 working paper SSRN 4198438 +37% is retired); Ravichandran & Korula (Google 2019, mark vendor-conflict); Bajari, Chernozhukov, Hortacsu & Suzuki (NBER WP 24334, 2018); de Fortuny, Martens & Provost (Big Data 2013); Varian (JEP 2014); Neumann, Tucker & Whitfield (Marketing Science 2019). All are VERIFIED-secondary except E46-E47 (VERIFIED-primary); E45 (GumGum) is UNVERIFIED and must not be cited as load-bearing.
- **A3:** M(u,y) form review (Section 7).
- **A0 / ledger:** the CANON-LEVEL entry proposed in draft-v1 Section 6 stands; this note adds the empirical grounding (the 678x/31,000x replacement is now anchored in primary filings, not literature summary). **Run-3 CANON-LEVEL escalation (L133; finalised at L134):** v2’s “aggregation wedge is super-additive, order 10^{1-105} ” does NOT survive the returns-to-scale evidence; the word “super-additive” is dropped entirely (no figure demonstrates it and E47 affirmatively rejects it in the variety dimension), what survives is a single-record modelling premium of order $\sim 10^0$ that saturates, and the large observer-to-broker gap is rent, not aggregation. This supersedes the wedge characterisation in L130/L132. The register process, not A12, decides the canon surface (GR-10).